

Advertising, Sales & Website Visits

Complete Covariance Calculation — No Shortcuts

Important: We calculate one variable at a time for variance, and two variables at a time for covariance. We never multiply all three columns together.

1. Dataset

Advertising_Spend	Sales	Website_Visits
10	105	1200
12	112	1350
15	120	1500
18	128	1680
20	135	1800
22	141	1950
25	150	2150
28	158	2300
30	165	2450
32	171	2600
35	180	2800
38	188	3000

2. Formulas

Mean = Sum / n

Sample Variance = $\sum (X - \bar{X})^2 / (n - 1)$

Sample Covariance = $\sum [(X - \bar{X})(Y - \bar{Y})] / (n - 1)$

3. Advertising_Spend Variance — Every Calculation

Step 1: Add all values:

$10 + 12 + 15 + 18 + 20 + 22 + 25 + 28 + 30 + 32 + 35 + 38 = 285$

Step 2: Count values: $n = 12$

Step 3: Mean = $285 / 12 = 23.75$

Step 4: Calculate $X - 23.75$, then square it.

X	$X - 23.75$	$(X - 23.75)^2$
10	-13.75	189.0625
12	-11.75	138.0625
15	-8.75	76.5625
18	-5.75	33.0625
20	-3.75	14.0625
22	-1.75	3.0625
25	1.25	1.5625
28	4.25	18.0625
30	6.25	39.0625
32	8.25	68.0625
35	11.25	126.5625
38	14.25	203.0625
TOTAL		834.4167

Step 5: $n - 1 = 12 - 1 = 11$

Variance = $834.4167 / 11 = 75.85416667$

Answer: 75.85416667

4. Sales Variance — Every Calculation

Step 1: Add: $105 + 112 + 120 + 128 + 135 + 141 + 150 + 158 + 165 + 171 + 180 + 188 = 1753$

Step 2: $n = 12$

Step 3: Mean = $1753 / 12 = 146.0833333$

Step 4: Calculate $Y - 146.0833333$, then square it.

Y	$Y - 146.0833333$	$(Y - \blacksquare)^2$
105	-41.0833	1687.8403
112	-34.0833	1161.6736
120	-26.0833	680.3403
128	-18.0833	327.0069
135	-11.0833	122.8403
141	-5.0833	25.8403
150	3.9167	15.3403
158	11.9167	142.0069
165	18.9167	357.8403
171	24.9167	620.8403
180	33.9167	1150.3403
188	41.9167	1757.0069
TOTAL		2475.9681

Variance = $2475.9681 / 11 = 225.0880101$

Answer from the supplied data: 225.0880101

The value 225.5208333 written in the question does not match these 12 Sales values.

5. Website_Visits Variance — Every Calculation

Step 1: Add: 1200 + 1350 + 1500 + 1680 + 1800 + 1950 + 2150 + 2300 + 2450 + 2600 + 2800 + 3000 = 24780

Step 2: n = 12

Step 3: Mean = 24780 / 12 = 2065

Step 4: Calculate Y – 2065, then square it.

Y	Y–2065	(Y–2065) ²
1200	-865	748225
1350	-715	511225
1500	-565	319225
1680	-385	148225
1800	-265	70225
1950	-115	13225
2150	85	7225
2300	235	55225
2450	385	148225
2600	535	286225
2800	735	540225
3000	935	874225
TOTAL		3812250

Variance = 3,812,250 / 11 = 346,568.1818

Answer from the supplied data: 346,568.1818

The value 310141.6667 written in the question does not match these Website_Visits values.

6. Advertising_Spend × Sales Covariance

Mean X = 23.7500000; Mean Y = 146.0833333; n = 12; n – 1 = 11

For every row: calculate $X - \bar{X}$, calculate $Y - \bar{Y}$, then multiply the two deviations.

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$
10	105	-13.7500	-41.0833	564.8958
12	112	-11.7500	-34.0833	400.4792
15	120	-8.7500	-26.0833	228.2292
18	128	-5.7500	-18.0833	103.9792
20	135	-3.7500	-11.0833	41.5625
22	141	-1.7500	-5.0833	8.8958
25	150	1.2500	3.9167	4.8958
28	158	4.2500	11.9167	50.6458
30	165	6.2500	18.9167	118.2292
32	171	8.2500	24.9167	205.5625
35	180	11.2500	33.9167	381.5625
38	188	14.2500	41.9167	597.3125
TOTAL				2706.2500

Covariance = 2706.2500 / 11 = 246.0227273

Supplied matrix value: 670.7430556 (does not match supplied data)

7. Advertising_Spend × Website_Visits Covariance

Mean X = 23.7500000; Mean Y = 2065.0000000; n = 12; n – 1 = 11

For every row: calculate $X - \bar{X}$, calculate $Y - \bar{Y}$, then multiply the two deviations.

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$
10	1200	-13.7500	-865.0000	11893.7500
12	1350	-11.7500	-715.0000	8401.2500
15	1500	-8.7500	-565.0000	4943.7500
18	1680	-5.7500	-385.0000	2213.7500
20	1800	-3.7500	-265.0000	993.7500
22	1950	-1.7500	-115.0000	201.2500
25	2150	1.2500	85.0000	106.2500
28	2300	4.2500	235.0000	998.7500
30	2450	6.2500	385.0000	2406.2500
32	2600	8.2500	535.0000	4413.7500
35	2800	11.2500	735.0000	8268.7500
38	3000	14.2500	935.0000	13323.7500
TOTAL				58165.0000

Covariance = 58165.0000 / 11 = 5287.7272727

Supplied matrix value: 14416.25 (does not match supplied data)

8. Sales × Website_Visits Covariance

Mean X = 146.0833333; Mean Y = 2065.0000000; n = 12; n – 1 = 11

For every row: calculate $X - \bar{X}$, calculate $Y - \bar{Y}$, then multiply the two deviations.

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$
105	1200	-41.0833	-865.0000	35537.0833
112	1350	-34.0833	-715.0000	24369.5833
120	1500	-26.0833	-565.0000	14737.0833
128	1680	-18.0833	-385.0000	6962.0833
135	1800	-11.0833	-265.0000	2937.0833
141	1950	-5.0833	-115.0000	584.5833
150	2150	3.9167	85.0000	332.9167
158	2300	11.9167	235.0000	2800.4167
165	2450	18.9167	385.0000	7282.9167
171	2600	24.9167	535.0000	13330.4167
180	2800	33.9167	735.0000	24928.7500
188	3000	41.9167	935.0000	39192.0833
TOTAL				172995.0000

Covariance = 172995.0000 / 11 = 15726.8181818

Supplied matrix value: 310141.6667 (does not match supplied data)

9. Final Covariance Matrix Calculated Directly From Your 12 Rows

	Advertising	Sales	Website
Advertising	82.7500000	246.0227273	5287.7272727
Sales	246.0227273	731.7196970	15726.8181818
Website	5287.7272727	15726.8181818	338336.3636364

Final check: each diagonal value is a variance. Each off-diagonal value is a covariance. The matrix is symmetric: $\text{Cov}(X,Y) = \text{Cov}(Y,X)$.

10. Easy Memory

Variance: ONE column → subtract mean → square → add → divide by $n-1$.

Covariance: TWO columns → find both means → subtract both means → multiply row by row → add → divide by $n-1$.